

Seeing Skin Cancer Before It's Too Late: The Rise of AI in Dermatology

Dermatology AI Interest Group, Monthly Spotlight: AI tools in Dermatology, skin cancer detection

The detection of skin cancer primarily relied on clinicians through visual inspection, going mainly through 2 steps: The eyes of an expert dermatologist, which either reassures you that it's benign and nothing to worry about or tells you to proceed with the next step by taking a biopsy and checking them by histopathologist to reach the promised diagnosis.

However, living in the world of technology, AI has stepped hugely into clinical diagnosis and strongly into the field of dermatology, evaluating the steps towards the right diagnosis by rate, efficiency, and reliability, changing the game in the field. The promise is straightforward: if an algorithm can flag which skin lesions are low risk and which warrant urgent attention, it could reduce unnecessary referrals for benign lesions while ensuring malignant ones are detected sooner. But promise and real-world performance are not the same thing, and the story of AI in skin cancer detection is really a story of that gap slowly narrowing, and of the problems that remain even as it does.

The Foundational Proof of Concept

The start of AI in skin cancer detection began not long ago; a study in 2017 published in Nature kicked off the use of AI tools in the field, which is now considered to be a landmark study in Nature. What made this study special than any other, rather than relying on manually selected image features, they trained a convolutional neural network (CNN) directly on raw clinical images, allowing the model to learn disease-specific visual patterns on its own. Using a dataset of 129,450 clinical images representing more than 2,000 skin diseases, this huge dataset has never been offered in any other study before, and the aim of this important study was to test the performance of a single deep learning model compared to a trained dermatologist. To test its performance, 2 common cases were presented in the study: differentiating malignant keratinocyte carcinomas from benign seborrheic keratoses and distinguishing malignant melanoma from benign nevi. The results were unexpected; across all 3 tasks, the CNN achieved an area under the

curve of 91% , and in many cases outperformed the performance of the board-certified dermatologist in comparison. This study fundamentally changed expectations for AI in dermatology. It demonstrated that a general-purpose deep learning model, trained solely on labeled images, could achieve expert-level diagnostic accuracy without handcrafted image features, shaping the next step of needed research in this field to answer an important question: could such performance be maintained outside carefully controlled laboratory settings?

Validating AI Under Real-World Conditions

Following Nature's study, research shifted its aim from checking the feasibility of using AI systems in the field to how well these systems are performing in more realistic environments. A study conducted in 2023 came to test 2 models, *Analyze* and *Detect*, using ordinary smartphone photographs rather than standardized dermoscopic images. As always done, the models' performance was compared to an expert dermatologist's performance. The results showed a high sensitivity, reaching 95.35% for *Analyze* and 96.4% for *Detect*, meaning they both were able to detect the benign from non-benign lesions, and very few lesions were missed and needed further assessment. Specificity was also high, reaching 90.32% and 94.85%, respectively. However, their performance declined when differentiating among malignant lesions themselves, reaching 72-76%, meaning the models couldn't accurately subclassify the malignant lesions. Nevertheless, the study provided a strong indication of the possibility of using ai models not only with dermoscopic images but also with smartphone images. Moreover, researchers evaluated the AI performance in larger, real-world clinical settings. The results were excellent, with both high sensitivity and specificity suggesting the use of AI tools in skin cancer screening; its role should be taken in consideration along with a clinician's judgment.

Inside the Clinic: AI x Dermatologists

Another question popped up in the research field: how can these models be used in actual routine clinical conditions, and what limitations do they offer?

An interesting observation emerged from a primary care pilot study that compared AI predictions with physician assessments. It showed that the AI model used helped in reducing unnecessary consultations for low-risk lesions, and also influenced the suggested intervention of the dermatologist to a better one in some cases. Notably, the limitations were the actual use of the AI model by the patients; thus, the technological barrier was addressed in this study. Another important finding from a SRMA published in *npj Digital Medicine* in 2024, synthesized evidence from ten studies comprising more than 67,000 individual diagnoses. Clinicians working without AI achieved a sensitivity of 74.8% and a specificity of 81.5% , and when supported by AI, these values increased to 81.1% and 86.1%, respectively. Importantly, the reference standard used in the pooled studies ranged from non-dermatologists to board-certified ones; thus, the less experienced practitioners benefited more than others. These findings support the idea of using AI alongside the assessment of an actual expert.

Patient Perspectives on AI-Assisted Skin Cancer Diagnosis

From the perspective of a patient, the technical performance is not the only concern to take into consideration, but the trust and the ease of use of these tools need to be integrated with the judgment of their performance. Overall, the public attitude towards using AI in this field was overwhelmingly positive, but the enthusiasm was accompanied by the same point suggested by researchers: AI should be used as a support decision-making tool and not replace the experts in the field.

Two Heads Better Than One? What the Evidence Adds Up To

If one consistent thing mentioned in all studies done on the idea of AI use in dermatology was the poorly trained AI models on dark skin type, which is the cause of the bias in their results, and the reason for not generalizing its result on all population, thus, Diverse Dermatology Images (DDI) dataset: the first publicly available, biopsy-confirmed image set spanning the full range of Fitzpatrick skin types was conducted and tested using the AI models that had studies run on them. Their diagnostic performance significantly dropped, confirming the right to be concerned about the equity in use when it comes to

considering these AI models. Additionally, the lack of transparency in how these AI tools actually work, which if considered, can increase trust in their use by both clinicians and patients for these tools.

Where are we now in this field ?

Several image-based dermatology AI tools have secured CE marking in Europe, but as of the most recent literature reviewed here, none had secured full FDA approval in the United States, and direct-to-consumer applications in particular often launch with limited standardized validation. A group like the International Skin Imaging Collaboration has already started to evaluate guidelines for future experimental research in the field, taking all challenges mentioned above into consideration, to reach good-quality research with reliable and generalizable results.

References :

1. Artificial Intelligence in Skin Cancer Detection: Consumer Apps, Clinical Practice, and Imaging Technologies (review).
2. Esteva et al. (2017). Dermatologist-level classification of skin cancer with deep neural networks. *Nature*.
3. New AI algorithms on smartphones to detect skin cancer in a clinical setting — A validation study (2023).
4. Artificial intelligence in mobile health for skin cancer diagnostics at home. *EClinicalMedicine* (2025).
5. Evaluation of an artificial intelligence-based decision support for the detection of cutaneous melanoma in primary care. *British Journal of Dermatology* (2024).
6. Krakowski et al. (2024). Human-AI interaction in skin cancer diagnosis: a systematic review and meta-analysis. *npj Digital Medicine*.
7. Artificial Intelligence in Skin Cancer Diagnostics: The Patients' Perspective (survey study).
8. Daneshjou et al. (2022). Disparities in dermatology AI performance on a diverse, curated clinical image set. *Science Advances*.
9. Integrating Artificial Intelligence in Dermatological Cancer Screening and Diagnosis: Efficacy, Challenges, and Future Directions (review).

